



STATUS REPORT of MP1 MPPL Kot TURBOCHARGER FAILURE ANALYSIS

Equipment Supplier	M/s Wartsila		
Purpose of Report:	Management Appraisal and Root cause Analysis	Problem Reported	Abnormal Sound observed during trial of MP-1
		Date of Report	22.03.2021

Section 1: Problem Statement

On 01.02.2021, during trial of MP-1 after pre summer maintenance and lube oil pump repair, abnormal sound was observed from turbocharger assembly. Engine was stopped immediately for further checking.

Section 2: Background

3.1 Equipment (MP1) Details:-

a. Engine:-

Model	:- W6L32
Make	:- Wartsila India Pvt. Ltd.
Serial No	:- 20712
Rated BHP	:- 3146
Rated RPM	:- 750
Bore * Stroke	:- 320*400
Fuel	:- Diesel
Lube Oil	:- Servo Marine 1040
T/C Make	:- Napier
T/C Model	:- NA 297

3.2 Maintenance/ Repair history of the Equipment (MP3):-

The last major overhaul (48000 hrs maintenance) of the engine was done at CRH 50929 carried out from 02.05.19 to 15.06.19. During Major Overhaul, Glass bead cleaning and dynamic balancing of Turbocharger was carried out. The failure occurred at cumulative running hours of engine of 57661 hrs.

Section 4: Investigation

On 03.02.2021, Turbocharger assembly was dismantled and following were the observations.

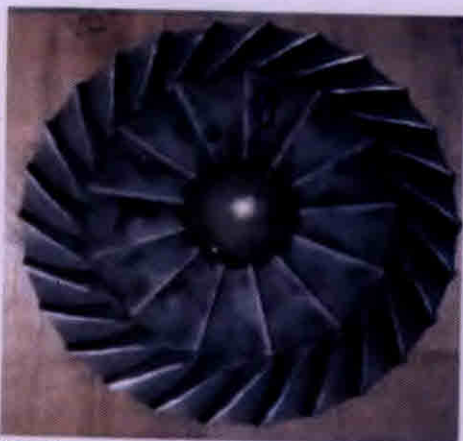
1. The compressor wheel had come in contact with the enclosure so there is damage on wheel. Vanes of compressor wheel found worn out throughout 360° i.e. vanes profile of



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compressor wheel was completely damaged due to heavy thrust applied from turbine side which is shown in picture.

2. The thrust collar and thrust plate have rubbed with each other so both are damaged.
3. The whole rotor assembly moved axially so the turbine side cone is damaged.
4. The C/E and T/E bearings are found ok with small hairline marks.
5. Lube oil pressure was found in lower limit and lube oil pump was also dismantled.
6. Bearings of lube oil pump were found worn out.

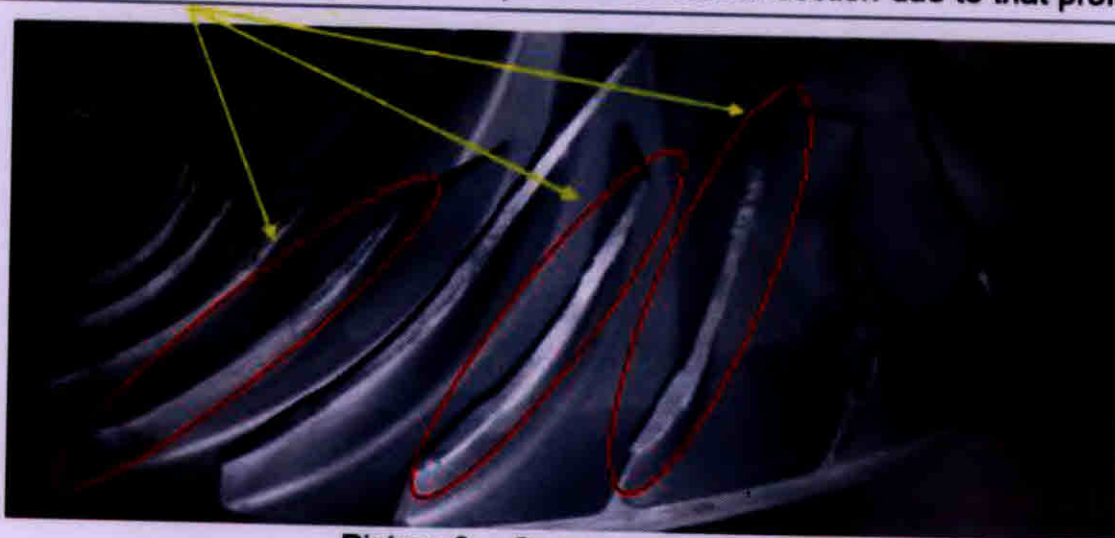


Picture 1: - Compressor wheel



Picture 2: - Compressor wheel

Vaness are worn out in between impeller and inducer section due to that profile



Picture 3: - Compressor wheel



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Section 5: Restoration

1. Compressor wheel, thrust collar and compressor cone were replaced.
2. Glass bead cleaning of Turbine and dynamic balancing of rotor assembly was also carried out.
3. Clearances were again maintained within prescribed limits and crosschecked.
4. Turbo assembly was again assembled and put into place.
5. Lube oil pump bearings were also replaced.
6. Engine trial was again taken 20.03.2021 and turbo was found in good condition.

Section 6: Conclusion

Preliminary investigation indicates that the failure may be because of lower lube oil pressure due to which bearing of turbocharger were overheated and hence could not bear the axial load.

The entire rotor assembly due to excess axial force deflected towards compressor side and hence compressor wheel and cone came in 360° contact and hence the markings and noise. Rest all related damage were collateral occurring after above damage.

Presently, Engine is available. WRPL, Kot has been advised to continuously monitor the fluctuation in Lube Oil pressure, SCADA trend of exhaust temperature and Lube oil consumption so that in case of any deviations, it may be noted and corrective actions may be taken.

Submitted please.


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अनुरक्षण प्रबन्धक
22/03/21

महाप्रबन्धक (अनुरक्षण)

WRPL kot may be advised to monitor lube oil pressure fluctuation, change air pressure, temp. trend.


23/03/2021

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